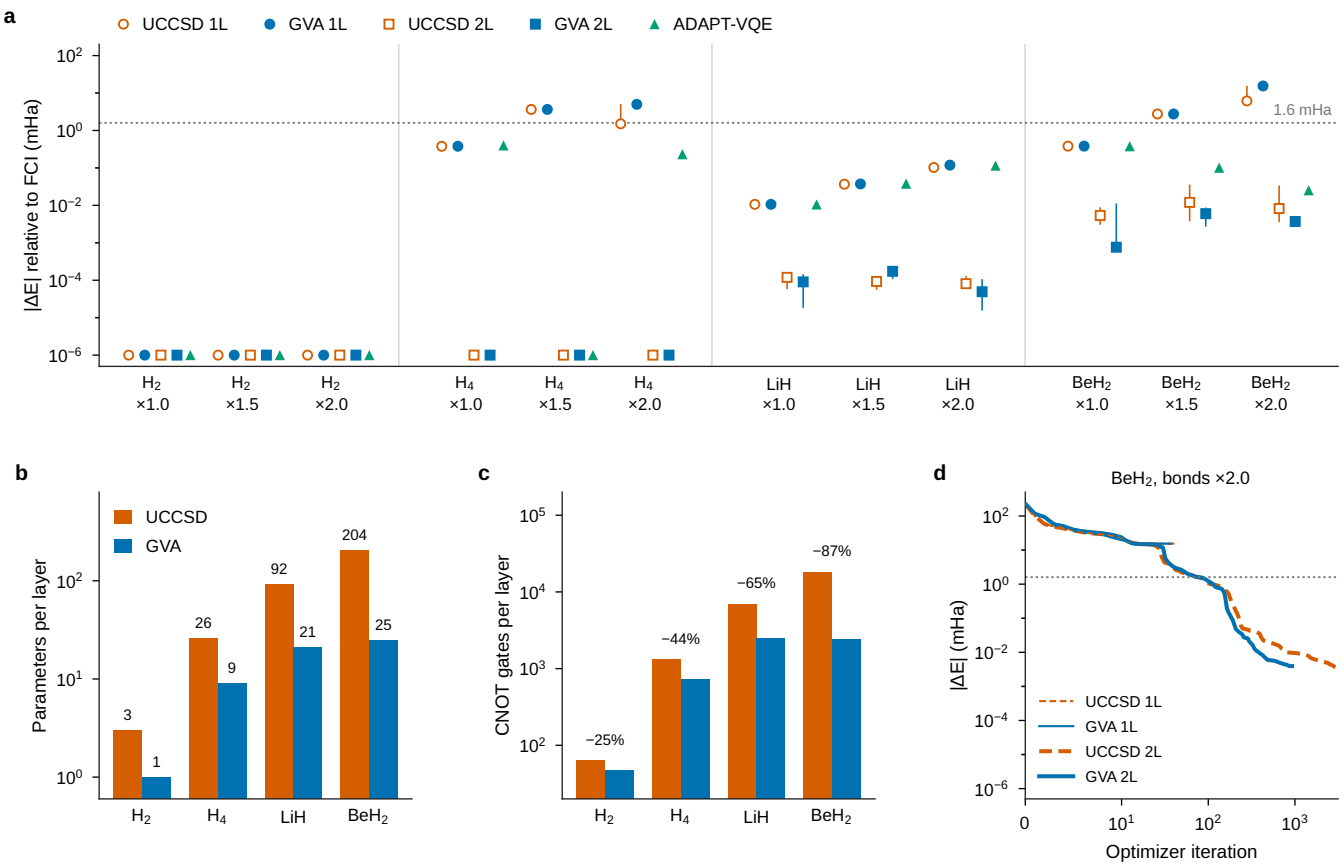
a
UCCSD 1L
GVA 1L
UCCSD 2L
GVA 2L
ADAPT-VQE
|ΔE| relative to FCI (mHa)
1.6 mHa
H2 ×1.0
H2 ×1.5
H2 ×2.0
H4 ×1.0
H4 ×1.5
H4 ×2.0
LiH ×1.0
LiH ×1.5
LiH ×2.0
BeH2 ×1.0
BeH2 ×1.5
BeH2 ×2.0
b
UCCSD
GVA
Parameters per layer
3
1
26
9
92
21
204
25
H2
H4
LiH
BeH2
c
CNOT gates per layer
−25%
−44%
−65%
−87%
H2
H4
LiH
BeH2
d
BeH2, bonds ×2.0
|ΔE| (mHa)
UCCSD 1L
GVA 1L
UCCSD 2L
GVA 2L
Optimizer iteration